

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

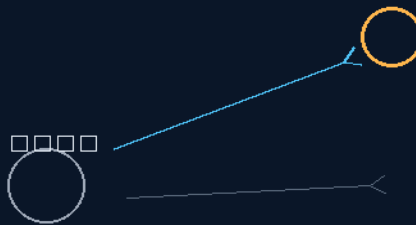
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 86

TARGETED SCRAP YARD LOGISTICS & SOLAR THERMONUCLEAR WASTE ELIMINATION

One yard per craft — tagged and priced at build
spent mass rides a one-way vector to the Sun



catalogue the scrap — feed the Sun

TAGGED AT BUILD · TARIFF PRICED · SUN AT END

CATALOGUE THE SCRAP — FEED THE SUN

Waste elimination — v1.0 in force

GP-IM-086

Revision v1.0 — 22 August 2026

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VETTED — GEMINI + KIMI + CHATGPT — 22 AUGUST 2026 — SHIP IT

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 86

PHASE 86: TARGETED SCRAP YARD LOGISTICS & SOLAR THERMONUCLEAR WASTE ELIMINATION —

STATUS: CURRENT. Orbit is a junkyard; the fleet makes it a warehouse. Phase 86 forces all corporate, private and sovereign satellite producers to pay a mandatory Scrap Ejection Tariff financing continuous orbital sweeping, and outlaws the intentional atmospheric burning or graveyard-orbit dumping of defunct platforms — the Kessler hazard answered with a ledger instead of a prayer. Galactic Version 2 shuttles and Phase 41 autonomous Mule Tugs capture and consolidate dead orbital mass; the scrap is fitted with Phase 15 directional actuators and steered into pinpoint landings on un-inhabited, zero-atmosphere moons, building high-density, fully mapped open-air repositories of refined aerospace titanium, copper and aluminum — the turn-back lifelines: a vessel with severe structural failure drops onto the scrap node and harvests refined stock to feed its Phase 85 foundries and CNC lathes, rebuilding critical systems without Earth resupply, ten-to-twenty years downstream. And the final vector: terrestrial high-level nuclear waste is permanently exhumed from burial vaults, sealed in hyper-armored Phase 22 Babushka pods, and propelled by the Phase 29 Mayernick electromagnetic runways into terminal ballistic trajectories directly into the Sun — the pods vaporizing in the photosphere, diluted into a star — what no planetary ecosystem should hold, gone for good. The vet grades the recycling architecture PASS and flags the disposal route for its own orbital/energy analysis — owed in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-086, v1.0 — 22 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the scrap law as seated (contents line, the two design bodies — verbatim) — §2 why it works (logistics and orbital mechanics — graded) — §3 the system in the yard — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 86: **Targeted Scrap Yard Logistics & Solar Thermonuclear Waste Elimination**. Establishes standard protocols for harvesting raw industrial scrap alloys from decommissioned installations, routing non-recyclable hazardous toxic byproducts straight into local stellar bodies — vaporized and diluted into the star, removed from the planetary ecosystem forever.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Kessler Syndrome Hazards & Subterranean Terrestrial Radioactive Burial [extracted verbatim from the combined masthead line]'

THE SOVEREIGN ORBITAL SCRAP EJECTION TARIFF (VERBATIM)

* Regulatory Mandate: Forces all corporate, private, and sovereign government satellite producers to pay a mandatory Scrap Ejection Tariff to finance continuous orbital sweeping. * The De-Orbit Prohibition: Outlaws the intentional atmospheric burning or graveyard-orbit dumping of defunct space platforms, boosters, and satellite electronics. * Automated Sweeper Sweeps: Deploys Galactic Version 2 shuttles and Phase 41 autonomous Mule Tugs to actively capture and consolidate all dead orbital mass.

EXOPLANETARY REPOSITORY DEPOTS (THE TURN-BACK LIFELINES) (VERBATIM)

* Precision Ballistic Vectoring: Captured orbital scrap is fitted with Phase 15 directional actuators and steered into pinpoint landings on un-inhabited, zero-atmosphere moons. * Pre-Processed Material Storage: Constructs high-density, fully mapped open-air scrap repositories consisting of pure refined aerospace titanium, copper, and aluminum alloys. * Decadal Emergency Intercepts: Serves as a 10-to-20 year downstream mission turnaround anchor. Vessels suffering severe, un-flight-mode structural failures can drop onto the scrap

node, harvesting refined raw materials to feed Phase 85 on-board foundries and CNC lathes to rebuild critical ship systems without Earth resupply. THE SOLAR THERMONUCLEAR ELIMINATION VECTOR * Terrestrial Waste Extraction: High-level nuclear waste arrays are permanently exhumed from terrestrial earth burial vaults and sealed inside hyper-armored Phase 22 Babushka pods. * Sub-Surface Burial Prohibition: Flatly outlaws the subterranean storage or industrial usage of depleted uranium and toxic chemical materials across all Earth-bound sectors. * Electromagnetic Solar Injection: Waste pods are propelled via Phase 29 Mayernick electromagnetic runways into terminal ballistic trajectories directly into the Sun, the pods vaporize and mix in the photosphere — nothing reaches the core, and no star switches off an isotope; what the Sun does is remove the waste from the planetary ecosystem permanently: gone, diluted into a star, never again part of any biosphere.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE RECYCLING ARCHITECTURE, AFFIRMED: recovering structural metals from discarded machinery and remelting/refining them is conventional materials recycling — the vet graded the core concept PASS. The novelty is jurisdictional and logistical: the tariff that funds the sweep and the prohibition that stops the bleeding.

THE KESSLER ANSWER, GRADED AS GOVERNANCE: orbital debris is a tragedy-of-the-commons problem; the Scrap Ejection Tariff prices the externality and the de-orbit prohibition closes the lazy exits. A ledger instead of a prayer.

THE REPOSITORY LOGIC, AFFIRMED: pre-positioned, mapped, refined stock on zero-atmosphere moons converts 'stranded' into 'supplied' — the 10-to-20-year turnaround anchor is the difference between a casualty and a repair job. The pinpoint ballistic vectoring owes its accuracy figures to the bench.

THE SOLAR VECTOR, GRADED HONESTLY: dropping mass into the Sun is energetically expensive — from Earth orbit it requires shedding nearly all of Earth's orbital velocity, which is why the vet flagged the disposal route for its own orbital/energy analysis. The Mayernick runway is the program's answer to that cost; the analysis is owed, and the phase does not pretend otherwise — the architecture stands on the recycling, the solar line stands as the terminal answer for the worst isotopes, awaiting its numbers.

§3 — THE SYSTEM IN THE YARD

- THE TARIFF: mandatory Scrap Ejection Tariff on all satellite producers — corporate, private, sovereign — financing continuous sweeping.
- THE PROHIBITION: no atmospheric burning, no graveyard-orbit dumping of defunct platforms.
- THE SWEEP: Galactic V2 shuttles and Phase 41 Mule Tugs capturing and consolidating dead orbital mass.
- THE DEPOTS: Phase 15 actuators steer the scrap to pinpoint landings on zero-atmosphere moons — mapped repositories of refined titanium, copper, aluminum; the turn-back lifelines for crippled vessels.
- THE TERMINAL VECTOR: exhumed terrestrial nuclear waste in Phase 22 Babushka pods, propelled by Phase 29 Mayernick runways into the Sun.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE LEDGER-NOT-PRAYER DOCTRINE (seated with the tariff): the commons is protected by pricing the externality — the polluter funds the sweep.

- THE TURN-BACK DOCTRINE (seated with the depots): every scrap node is a lifeline — the junk of the last decade is the repair stock of the next.
- THE NO-BURIAL DOCTRINE (seated with the solar vector): what no ecosystem should hold does not go back into the ground — it goes into the furnace that can actually finish it.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Charter the tariff: the Scrap Ejection Tariff drafted into the orbital regulatory framework; the sweep budget defined.
- STEP 2 — Fly the sweep: Mule capture drills on cooperative dead-mass targets; consolidation and actuator fitting rehearsed.
- STEP 3 — Bench the vectoring: pinpoint-landing accuracy of actuator-steered scrap on the depot approach — the figures published.
- STEP 4 — Open the first depot: a mapped repository site; inventory schema (alloy, form, mass, location) live.
- STEP 5 — Drill the turn-back: a simulated crippled vessel harvesting depot stock through the Phase 85 foundry chain to a finished part.
- STEP 6 — Number the solar vector: the orbital/energy analysis of the Sun-bound trajectory from the Mayernick runway — the vet's owed item — published before any waste pod flies.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 41 — the Mule Tugs (GP-IM-041): the sweep fleet.
- PHASE 15 — the directional actuators (GP-IM-015): the vectoring hardware.
- PHASE 85 — the vetting engine and foundries (GP-IM-085): the consumer of the depot stock.
- PHASE 22 — the Babushka pods (GP-IM-022): the waste containment.
- PHASE 29 — the Mayernick runways (GP-IM-029): the solar-injection propulsion.
- PHASE 33 — the moon depots (GP-IM-033): the repository family.

§8 — DO-NOT-BUILD

- NEVER burn or graveyard a platform — the prohibition is absolute; the sweep exists so the lazy exits

stay closed.

- NO unmapped stock — a repository without the inventory schema is a junkyard; the map is the difference.
- NEVER fly a waste pod before the analysis — the solar vector's orbital/energy budget publishes first (the vet's owed item).
- NO burial — subterranean storage of the worst isotopes is outlawed across all Earth-bound sectors; the ground is not a solution.

§9 — OWED

THE ANALYSIS, OWED (the vet's item): the orbital/energy budget of the solar-injection trajectory from the Mayernick runway; the vectoring accuracy figures for depot landings; the turn-back drill report.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-086, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 22 August 2026, ninth of the new 'next 10' shift ('10 more before go to the old lady for dinner'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562 and 564): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 86